

I Support Intelligence & Cancellation Analysis |

An End-to-End Customer Churn Analytics Workflow

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customer_churn_data_raw.xlsx

Core Technologies Used:

Python 3

SQLite

Pandas & NumPy

Matplotlib & Seaborn

Introduction

Customer churn is an important business problem because losing customers can directly affect revenue and long-term business growth. This project focuses on analysing customer churn data to understand why customers cancel their subscriptions and which customer segments show higher churn.

The project uses Python, Pandas, NumPy, Matplotlib, Seaborn, and SQLite for data storage, cleaning, analysis, and visualization. Customer, subscription, and support data are combined into a single dataset for analysis.

The analysis covers churn rate, retention rate, churn by plan type, churn by state, churn by subscription type, customer age, customer tenure, revenue at risk, complaints, escalations, and churn risk. The objective is to convert raw customer data into useful business insights that can support customer retention strategies.

Learn-Support Intelligence and Cancellation Analysis

1. The Business Challenge In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You will act as a Data Analyst tasked with identifying high-risk subscribers using a multi-dimensional dataset (Customer demographics, Subscription tiers, and Support escalations).
2. Core Tech Stack SQL & Python Integration (numpy, pandas, sqlite3, matplotlib, seaborn), Performance Engineering (data cleaning, feature engineering, analytics), Behavioral Visualization, Writing Actionable Insights
3. Project Milestones
 - Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets.
 - Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.
 - Executive Reporting: Translating technical findings into billion-dollar business insights.

Churn Analysis - Roadmap

- Connect SQL database to Python – pandas and sqlite3
- Data import using SQL query in Python – pandas and sqlite3
- Data Cleaning – numpy and pandas
- data types, rename cols, select specific cols, QCs, handle missing/null values
- Feature Engineering – numpy and pandas
- Create new calculated cols, data transformation, use filters
- Data Analysis – numpy and pandas
- EDA - aggregation, group by, pivot table
- Data Visualization – matplotlib and seaborn

Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you.

For this, we only need to look at three things:

- **"Who"**: Identifying which customers left or will leave in future
- **"Why"**: Analysing their behavior before they left.
- **"When"**: Finding the "danger zone" – point when a user leave

Business Type	Churn Definition
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Using service/app in 90 – 120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

Database Tables

Database: customer_churn



db_customer
customerid
name
country
State
gender
dob
interests
pincode



db_subscription
customerid
subscription_start_date
subscription_type
renewal_date
plan_type
contract_type
cancellation_date
cancellation_reason
monthly_charges
cltv
churn_score



db_support
customerid
complaint_date
Escalations
csat_score
col_1
comment

Calculated metrics

KPI	Formula
Churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1 - churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100
Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

Insights

Overall Churn

The overall churn rate is **28.57%**.

The retention rate is **71.43%**.

Churn by Plan Type

Basic plan has a **60.00% churn rate**.

Standard plan has a **22.22% churn rate**.

Premium plan has a **14.29% churn rate**.

Revenue at Risk

Monthly charges associated with churned customers total **73.94** in the project's reported revenue-at-risk measure.

Customer Tenure

Average customer tenure is approximately **1,550 days**.

Support Escalations

The escalation rate is **19.05%**.

The project reports a **0.77 correlation** between escalation and churn flag. This indicates a strong positive association within this dataset; it does not by itself establish that escalations cause churn.

Customer Complaints

The average number of complaints per user is **0.43**.

Churn Risk

Customers are categorized into low, medium, and high churn-risk groups using their churn scores.

Scores below 50 are classified as low risk, 50–69 as medium risk, and 70 or above as high risk.

Subscription Type

Among churned users, the analysis shows **Paid** and **Referral** subscription types, with the reported revenue and user counts shown in the project output.

Project Summary

This project provides an end-to-end analysis of customer churn using customer, subscription, and support data. The data was first stored and managed using SQLite and then cleaned and analyzed using Python.

The project identified important customer and business metrics including a **28.57% churn rate**, **71.43% retention rate**, **18.85 average revenue per user (ARPU)**, approximately **1,550 days average tenure**, and **73.94 revenue at risk** based on monthly charges from churned customers. The analysis also examined churn across plan types, states, subscription types, customer age, tenure, complaints, escalations, and churn-risk categories. Visualizations using Matplotlib and Seaborn were used to make these patterns easier to understand.

Overall, the project demonstrates how raw customer data can be transformed into structured analysis and business insights. The findings can help a business monitor customer churn, identify higher-risk segments, understand potential revenue exposure, and focus retention efforts on areas where churn is more prevalent.